Carson Kohlbrenner

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RESEARCH SUMMARY

Second year Ph.D. student researcher in robotics at the University of Colorado Boulder specializing in embodied learning, robot world models, and multimodal perception. Developed world models for humanoid collision avoidance using large-scale simulation and distributed GPU training, alongside first-author ICRA research on whole-body tactile and proximity sensing. Experience spanning robot learning, reinforcement learning, simulation, and physical robotic systems.

EDUCATION

Ph.D. in Robotics - University of Colorado Boulder — Aug 2025 – Current
Department: Computer Science

M.Sc. in Autonomous Systems - University of Colorado Boulder — Aug 2024 – May 2025
GPA: 4.0/4.0
Department: Ann & H.J. Smead Aerospace Engineering Sciences
Thesis Title: "Procedurally Generated Whole-Body Artificial Skin for Robot Applications"

B.Sc. in Aerospace Engineering - University of Colorado Boulder — Aug 2020 – May 2024
GPA: 3.9/4.0
Minor: Computer Science. *Awards:* Summa cum laude

RELEVANT RESEARCH: HUMANOID WORLD MODELS

Note: 'C' = conference paper and 'W' = workshop paper. All works have been accepted at peer-reviewed venues except those currently still in review, denoted by [In Review].

[C, 2027] **C. Kohlbrenner**, N. Pudasaini, W. Xie, N. Sivagnanadasan, N. Correll, A. Roncone, "**Move, don't think: sample-efficient collision avoidance with lossy proximity sensors**" in *International Conference on Learning Representations (ICLR)*. [In Review]

- Developed and pretrained a world model for Unitree H1-2 humanoid collision avoidance using 6,000 expert demonstrations simulated in Isaac Lab and 64 distributed egocentric depth sensors.
- Reconstructed anticipated 4D environmental geometry from predicted depth observations to study contact-aware robot policies.
- Optimized multi-day model pretraining on a cluster of NVIDIA H200 GPUs and evaluated the effect of world-model anticipation on downstream reinforcement learning.
- Demonstrated that contact anticipation can introduce a learning-efficiency tradeoff, slowing policy training by up to 2.3× in the studied setting.

RELEVANT RESEARCH: MULTIMODAL ROBOT LEARNING

[C, 2026] **C. Kohlbrenner**, A. Soukhovei, C. Escobedo, A. Roncone, "**Design, Mapping, and Contact Anticipation with 3D-printed Whole-Body Tactile and Proximity Sensors**" in *IEEE International Conference on Robotics and Automation (ICRA)*.
Outstanding Paper Award - CU Boulder

[C, 2025] **C. Kohlbrenner**, C. Escobedo, S. Bae, A. Dickhans, A. Roncone, "**GenTact Toolbox: A Computational Design Pipeline to Procedurally Generate Context-Driven 3D Printed Whole-Body Artificial Skins**" in *IEEE International Conference on Robotics and Automation (ICRA)*.

- Developed a whole-body tactile and proximity artificial skin sensor that automatically generates around a given robot body.
- Designs are 3D-printable and can be optimized with task-relevant data in Isaac Sim.
- Pretrained a multimodal cross-attention transformer on real camera, infrared depth, capacitive proximity, and magnetic tactile data to create a continuous sensing space for safe human-robot interaction.
- Developed an online ROS2 inference stack for contact-rich control and foundation model querying.

ALL PUBLICATIONS

[C, 2027] **C. Kohlbrenner**, N. Pudasaini, W. Xie, N. Sivagnanadasan, N. Correll, A. Roncone, "**Move, don't think: sample-efficient collision avoidance with lossy proximity sensors**" in *International Conference on Learning Representations (ICLR)*. [In Review]

[C, 2026] **C. Kohlbrenner**, A. Soukhovei, C. Escobedo, A. Roncone, "**Design, Mapping, and Contact Anticipation with 3D-printed Whole-Body Tactile and Proximity Sensors**" in *IEEE International Conference on Robotics and Automation (ICRA)*.
Outstanding Paper Award - CU Boulder

[C, 2026] H. Chen, **C. Kohlbrenner**, J. Kubik, L. Rustler, A. Dickhans, K. Bartunek, A. Roncone, H. Lee, M. Hoffmann, "**Toward Geometry-Scalable Whole-Body Touch for Humanoids: A 3D-Printed Conformal EIT Skin**" in *IEEE-RAS International Conference on Humanoid Robots (Humanoids)*.

[C, 2025] **C. Kohlbrenner**, C. Escobedo, S. Bae, A. Dickhans, A. Roncone, "**GenTact Toolbox: A Computational Design Pipeline to Procedurally Generate Context-Driven 3D Printed Whole-Body Artificial Skins**" in *IEEE International Conference on Robotics and Automation (ICRA)*.

[W, 2026] **C. Kohlbrenner**, N. Pudasaini, W. Xie, N. Sivagnanadasan, N. Correll, A. Roncone, "**Contact anticipation can reduce collision avoidance sample-efficiency**" in the *Do Robots Need World Models workshop at the Conference on Robot Learning (CoRL)*.

[W, 2026] **C. Kohlbrenner**, N. Pudasaini, W. Xie, N. Sivagnanadasan, N. Correll, A. Roncone, "**Egocentric Tactile and Proximity Sensors as Observation Priors for Humanoid Collision Avoidance**" in the *8th RoboTac Workshop at the IEEE International Conference on Robotics and Automation (ICRA)*.

[W, 2026] **C. Kohlbrenner**, C. Escobedo, S. Ray, A. Dickhans, A. Soukhovei, N. Jackoski, L. Antieau, A. Roncone, "**Improving Sensing Coverage and Compliance of 3D-Printed Artificial Skins Through Multi-Modal Sensing and Soft Materials**" in the *Towards Large-Area Tactile Sensing Skins workshop at the International Conference on Robotics and Automation (ICRA)*.

[W, 2026] H. Chen, **C. Kohlbrenner**, J. Kubik, L. Rustler, A. Dickhans, A. Roncone, H. Lee, M. Hoffmann, "**A 3D-Printed Electrical Impedance Tomography Tactile Sensor for Scalable Coverage**" in the *Towards Large-Area Tactile Sensing Skins workshop at the International Conference on Robotics and Automation (ICRA)*.
Best Poster Award

[W, 2025] **C. Kohlbrenner**, M. Murray, Y. Zhang, C. Escobedo, T. Dunnington, N. Stevenson, N. Correll, A. Roncone, "**A Machine Learning Approach to Contact Localization in Variable Density Three-Dimensional Tactile Artificial Skin**" in the *Workshop on Touch Processing at the Conference on Neural Information Processing Systems (NeurIPS)*.

[W, 2025] A. Soukhovei, **C. Kohlbrenner**, C. Escobedo, A. Roncone, "**Form-Fitting, Large-Area Sensor Mounting for Obstacle Detection**" in the *Workshop on Advances in Contact-Rich Robotics (ConRich) at IEEE-RAS International Conference on Humanoid Robots (Humanoids)*.

RELEVANT SKILLS

- **Programming and ML**: Python, C++, PyTorch, CUDA, Hugging Face
- **Robotics and Simulation**: Isaac Lab, Isaac Sim, ROS2, cuRobo, Gazebo, OpenCV
- **Robot Learning**: Reinforcement learning (PPO, SAC), model-based and model-free RL, VAE-FM world models, VLA/WAM architectures
- **Generative Modeling**: Transformers, VAEs, conditional flow matching, video generation
- **Large-Scale Training**: Multi-GPU training, DDP, FSDP, tensor/pipeline parallelism
- **Robotic Platforms**: Franka FR3 (real and sim), Franka Panda (real and sim), Unitree H1-2 (real and sim), Unitree Go2 (sim)

WORK EXPERIENCE

Research Assistant – Human Interaction & Robotics Group, Boulder, CO — May 2024 – Current
Description: Funded for a range of projects including building world models, novel tactile sensors, robotic chemistry, and autonomous drone inspection.

Teaching Assistant – University of Colorado Boulder, Boulder, CO — Jan 2024 – May 2026
Description: Facilitated undergraduate labs for the following courses: ASEN 3801 - Orbital Dynamics and Controls, ASEN 4018/4028 - Aerospace Senior Projects, and CSCI 1300 - Introduction to Computer Science.

Senior ML / Robotics Engineer – Foxglove, Remote — July 2025 – Jan 2026
Description: Created robotics demos that showcased Foxglove's data visualization software. Wrote tutorial articles and held workshops to teach customers how to use the Foxglove SDK for rapid data visualization.

Assistant Engineer/Technician – Technology Applications Inc., Gunbarrel, CO — July 2021 – Jan 2024
Description: Aided in the design and manufacturing of high-temperature thermal transfer straps for NASA and Army-funded SBIR programs. Performed and managed two full-company audits. Heavily involved in proposal writing.

AWARDS

Best Poster Award, ICRA Tactile Sensing Workshop — June 2026
Outstanding Paper Award, University of Colorado Boulder — Mar 2026
Summa cum laude, University of Colorado Boulder — May 2024
Sewall Esteemed Scholars Recipient, University of Colorado Boulder — Aug 2020

AGENTIC CODING STATEMENT

Primary coding workflow includes implementing research critical components by hand, while using coding agents to implement boilerplate features, discover sources, and answer questions about the code base. I write my own validation tests to verify any output that could modify research assumptions.